

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EC)

November – December 2011

EC 204 Electronics Instruments & Measurements

Date: 02.12.2011, Friday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a)

[04]

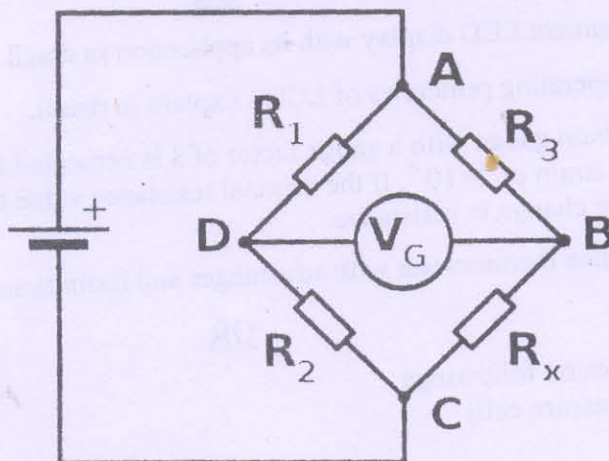
- 1) What is limiting errors or Guarantee errors?
- 2) Define the following terms:
 - a. Average value
 - b. Arithmetic mean

(b) Which bridge is used to measure frequency and explain the measurement procedure?

[03]

Q - 2 (a) The impedance of the basic AC bridge of below figure are given as

[03]



$R_1 = Z_1 = 100 \Omega$ with 80° (inductive impedance)

$R_3 = Z_2 = 250 \Omega$ (Pure Resistance)

$R_2 = Z_3 = 400 \Omega$ with 30° (inductive impedance)

$R_X = Z_X = \text{Unknown}$

Determine the constant of the unknown arm.

(b) Compare Hay's bridge with Maxwell bridge.

[03]

(c) 1) Explain the operation of period measurement with the multiple average mode.

[06]

2) What are the techniques used to reduce the input frequency (High frequency)?

[02]

OR

Q - 2 (a) Explain the working of Schering bridge.

[03]

- (b) With neat diagram explain in detail about Hay bridge. [03]
- (c) Explain the operation of a basic digital multimeter & How current and resistance can be measured in digital multimeter? [08]

- Q - 3 (a) Explain with the help of a neat diagram the working of a DFM. [04]
- (b) Explain dual trace oscilloscope. What are the advantages of dual trace over dual beam CROs for multiple trace? [06]
- (c) List out the applications of CRO and explain them in detail. [04]

OR

- Q - 3 (a) What is the function of a time base selector? How does it operate? [04]
- (b) Write a short note on digital storage oscilloscope (DSO). [06]
- (c) What are the major components of a CRT? [04]

SECTION – II

- Q - 4 (a) Explain fundamental suppression harmonic distortion analyzer. [07]
- Q - 5 (a) Write a short note on segmented gas discharge displays. [07]
- (b) How BCD to seven segment converters and BCD to dot matrix converter work? [07]

OR

- Q - 5 (a) Explain a 7 segment LED display with its application in detail. [07]
- (b) What are the operating principles of LCD? Explain in detail. [07]
- Q - 6 (a) A resistance strain gauge with a gauge factor of 3 is cemented to a steel member, which is subjected to a strain of 2×10^{-6} . If the original resistance value of the strain gauge is 150 Ω , calculate the change in resistance. [07]
- (b) Explain resistance thermometer with advantages and limitations. [07]

OR

- Q - 6 (a) Write short notes on followings [08]
- Load cell (Pressure cell)
 - Photo cells
- (b) Explain RTD and thermocouple in brief. [06]
